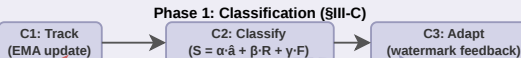


OrchKvCache Scheduler — C8: Tiered Manager (§III-G)



HOT / WARM / COLD

Phase 2: Eviction + Phase 3: Prefetch

C4: Evict
(coldest-first)

C5: Prefetch
(EMA max-heap, $K=8$)

C6: Pipeline
(overlap tracking)

C7: Migration Engine (§III-E)

demote
(GPU → DRAM → SSD)

promote
(SSD → DRAM → GPU)

Tier 0: GPU HBM
HOT | 23 GB/s

Tier 1: Host DRAM
WARM | 23 GB/s

Tier 2: SSD (OrchFS)
COLD | 17.8 / 5.3 GB/s

Tiered Storage Pools (§III-D)

Inference Engine

(HuggingFace / vLLM)

user
request

Prefill Phase

Decode Loop

Attention(Q, K, V)

Output Token

past_kv

